Ex. 2

30/07/2026

Study of Different Types of Network Cables.

Aim:

Understanding different types of network cables in Computer networks

Cable types	Category	Max data Transmission	Advantages/ Disadvantages	Uses.
UPT	Category 3	10 MbPS.	Advantages: → cheaper → easy to install as they have small diameter. Disadvantages: More Prone to EMI & Noise.	10 Base-T Ethernet.
	Category 5	upto 100MbPS		fast ethernet.
	Category 5e	1GbPS.		Gigabit. Ethernet.
STP.	Category 6, 6a.	10 GbPS.	Advantages: → Shielded → faster than UTP. → less Susceptible to noise. Disadvantage → Expensive → Greater installation effort.	Widely used in data centers
SSTP.	Category 7	10 Gbps		Gigabit ethernet (100m)

Cable type	Category	Max data Transmission.	Advantages/ disadvantages	Usages.
Coaxial Cable	RG-6 RG-59 RG-11	10 - 100 Mbps	Advantages. → High bandwidth → Immune to interference → low loss bandwidth → versatile disadvantage → limited distance → cost → Size is bulky.	Speed of signal is 500M. Television network. high speed network connection
fiber optics cable	Single mode Multi mode	100 Gbps	Advantages → High Speed → High bandwidth → high Security → long distance disadvantages: → Expensive → requires skilled installers.	Maximum distance of fibre optics cable is around 100M.

Result:

Thus, we have learned the different types of cables used in computer network successfully.